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Plant AI in Agriculture: Innovative Approaches to Sunflower Leaf Disease Detection with Federated Learning CNNs

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***Abstract*— This research presents a new use of Convolutional Neural Networks (CNN) integrated with Federated Learning (FL) for detecting and classifying illnesses on sunflower leaves. The centre of this investigation is a systematic analysis of findings obtained from local data, which are transformed into global knowledge with the help of sophisticated federated averaging approaches. The findings show the efficiency of our model, which is outstanding. The macro averages for accuracy, recall and F1-scores also reveal an upward trend of continuous improvement starting kr_1 (82.95) to up between 83.74 as at kr_3 met kp_4atkr(translation truncated). Weighted and micro averages also demonstrate a rising tendency, reflecting that the model becomes more accurate in terms of its ability to manage class imbalances. Specifically, the micro averages also improve similarly, starting at 83.05 and ending at 94.88 for kr_5. Still, the weighted ones change from just two decimal points below (from an average of close to 17 times ten minus six). That difference changes drastically because now we are going from having about. The research finds that federated averaging helps transform local data into global data. It demonstrates how the individual client can contribute individually to the learning of a model. Still, it also shows that in federated learning, collaborative and distributed nature has enhanced performance over time. In conclusion, the study demonstrates a convincing case for utilizing FL and CNN in rural situations, particularly for recognizing illnesses afflicting sunflower leaves. The experimental and statistical data reveal the efficiency of this method, which may lead to its broad utilization in plant pathology for better precision agriculture.**

Keywords: Sunflower leaves, Leaf diseases, (CNN)_(FL), Disease, Augmentation image, data privacy, data security.

I. INTRODUCTION

The area of machine learning, particularly the combination of Federated Learning (FL) and Convolutional Neural Networks (CNN), ushers in a new era in plant disease identification and classification[1]. The present study investigates how FL can be creatively combined with CNN to serve as a machine-learning algorithm for sunflower farming, one of the essential crops in Indian and global agriculture. Sunflowers have been valued for their oil and seeds but may be infected by various leaf diseases that considerably lower quality and production[2]. It is essential to treat these diseases quickly for economic sustainability and agricultural efficiency, especially where sunflower oil consumption remains a staple diet, such as in India. The main focus of this study is to identify and categorise the five main groups of diseases that affect sunflower leaves: powdery mildew, downy mildew, verticillium wilt rust, and botrytis. Given their diverse symptom manifestations, such an approach must be intricate and sophisticated for an accurate diagnosis[3]. This work is a fresh endeavour in which the classification of disease responsibility is divided among five customers, each with information set to show the different stages and heights of those illnesses. Besides enhancing the robustness of the illness detection model, this distributed approach eliminates possible privacy and data security issues usually associated with centralised systems[4]. This work is unique because it looks at more complex FL averaging methods that maximise the sum of learning models created by different customers. This approach results in a more accurate and generalizable model by ensuring the group's learning is more significant than its constituent subgroups[5]. Moreover, as the model uses CNNs—known for their robust visual recognition and classification abilities—it will certainly be able to detect patterns and unevenness on leaf images, which are an essential part of disease diagnostics. This study contributes to the practical answer to an unprecedented global and local dilemma that transcends into working theory[6]. In India, where millions of people rely on sunflower cultivation for survival and as a source of income, leaf diseases may lead to terrible consequences[7]. This cutting-edge, privacy-preserving technology, which this research presents as a handbook for increasing the performance of disease diagnosis, increases productivity in agriculture and farmers' financial stability. By marrying the capabilities of CNN and FL, this study intends to change how sunflower leaf illnesses are identified and grouped. This study is advanced by the desire to utilise sophisticated technical innovation to address age-old agricultural problems. The objectives might be outlined as follows: To Create a Sturdy CNN Model for Classifying Diseases: The project aims to design and implement a CNN model that effectively classifies five main categories of diseases affecting sunflower leaves[8]. This includes botrytis, downy mildew, powdery mildew, verticillium wilt, and rust. Every class has a unique clinical symptom; therefore, an intricate model that can detect slight patterns and differences in leaf photographs is needed[9]. To Put Federated Learning into Practice for Distributed Data Processing: The study aims to process data for five customers by leveraging FL's distributed nature[10]. This objective is crucial for diversifying the model data and ensuring comprehensive and overall learning[10][11]. FL's embedded privacy-preserving aspect is beneficial since it allows data sharing without losing confidentiality. To investigate and apply sophisticated federated averaging techniques: One of the main objectives is to analyse and utilise advanced averaging techniques for FL[12]. These approaches are crucial for enhancing the decentralized learning synthesis, ensuring that the integrated model beats separate client models' accuracy and generalizability limitations. To assess the effectiveness of the model in various agricultural environments: The proposed model's practical applicability and efficacy in varying agricultural conditions will be critically analysed, focusing on regions like India, where sunflowers are the primary crop. This study has several motivations, including a deep-seated dedication to tackling agricultural difficulties and technical curiosity. Agricultural Technology Advancement: A desire to push the boundaries of agricultural technology is front and centre. The main objective of this project is to highlight how contemporary machine-learning techniques could be applied to practical agrarian issues. Enhancing Crop Production and Quality: The research aims to significantly improve the yield and quality of sunflower harvests by accurately identifying and classifying leaf diseases. This is particularly vital for countries such as India, where sunflowers are a significant food crop. Empowering Farmers with Technology: The aim of the project is to make technology that can enable farmers, mainly but not exclusively in developing countries, to identify diseases early enough, thereby reducing crop losses and, in the end, promoting economic stability. Supporting Sustainable Agricultural Practices: Considering the urgent environmental concerns of our world, this study contributes to the broader goal of sustaining agriculture by enhancing more accurate and efficient disease management. Lastly, the potential of creating revolutionary Federated Learning (FL) use cases in the agriculture sector gives birth to a standard for the effective adoption of decentralized machine learning as well as real-world solutions. In other words, our research seeks to link sustainability in agriculture and technological innovation with the advent of novel machine-learning techniques.

II. Literature Review

Federated Learning (FL) and Convolutional Neural Networks (CNN) in agriculture: a relatively young yet rapidly evolving area that identifies and classifies sunflower leaf diseases[13]. This literature review views the current state of this research area, emphasising FL with CNN, which allows better detection of illness over remote datasets. Convolutional Neural Networks for Plant Disease Detection: CNNs have a long history of performance when identifying and classifying images[14]. In authors. published a pioneering study that revealed the power of CNNs in detecting plant diseases and how intelligent these networks can be at finding complex patterns emerging from leaf photographs. This work sets the precedent for the new use of CNNs in agriculture. Federated Learning in Agriculture: the researchers developed a model of FL that created the basis of their decentralised machine learning method, enabling it to offer an agricultural answer related to data privacy issues and heterogeneous needs for various information types. Jones and Patel validated distributed learning models in agricultural settings by leveraging FL to predict crop diseases. Advanced Federated Averaging Techniques: As the authors show, federated averaging methods have significantly developed[15]. Their initiative was focused on simplifying FL's aggregation process so that all client datasets with unique specifications could be added to the generalised model[5]. This approach is particularly relevant to our study, as it addresses the challenge of integrating information from different customers who each have a data set that reflects certain stages of illness on sunflower leaves. Classification of Sunflower Leaf Diseases: Though limited, studying individual sunflower leaf diseases has been instructive. For instance, the authors laid the groundwork for further studies by classifying sunflower diseases through standard machine-learning methods[16]. This work is unique because it looks at more complex FL averaging methods that maximise the sum of learning models created by different customers. This approach results in a more accurate and generalizable model by ensuring the group’s learning is more significant than its constituent subgroups. Moreover, as the model uses CNNs—known for their robust visual recognition and classification abilities—it will certainly be able to detect patterns and unevenness on leaf images, which are an essential part of disease diagnostics. This study contributes to the practical answer to an unprecedented global and local dilemma that transcends into working theory. In India, where millions of people rely on sunflower cultivation for survival and as a source of income, leaf diseases may lead to terrible consequences. This cutting-edge, privacy-preserving technology, which this research presents as a handbook for increasing the performance of disease diagnosis, increases productivity in agriculture and farmers’ financial stability. By marrying the capabilities of CNN and FL, this study intends to change how sunflower leaf illnesses are identified and grouped However, what distinguishes our study is the novel approach we propose for fusing CNNs for this aim, which ensures higher accuracy and efficiency. Difficulties in Indian Sunflower Cultivation: The role of sunflowers in Indian agriculture cannot be exaggerated[17]. The researchers analysis of the effect of leaf diseases on sunflower production in India shows that more advanced detection technologies are needed. This research provides our study with a comparative framework. It emphasizes the need to find a solution that was not only ahead of its time technologically but also economically feasible to work in various climatic and agricultural environments[18]. By marrying the capabilities of CNN and FL, this study intends to change how sunflower leaf illnesses are identified and grouped. This study is advanced by the desire to utilise sophisticated technical innovation to address age-old agricultural problems. The objectives might be outlined as follows: To Create a Sturdy CNN Model for Classifying Diseases: The project aims to design and implement a CNN model that effectively classifies five main categories of diseases affecting sunflower leaves [19].

III. METHODOLOGY

This research article systematically investigates the meat of a Fl framework and CNN combined with decision trees for identifying illnesses affecting sunflower leaves. This section provides the proposed approach consisting of a federated learning setup, model architecture, dataset preparation and evaluation, as demonstrated in Figure 1.

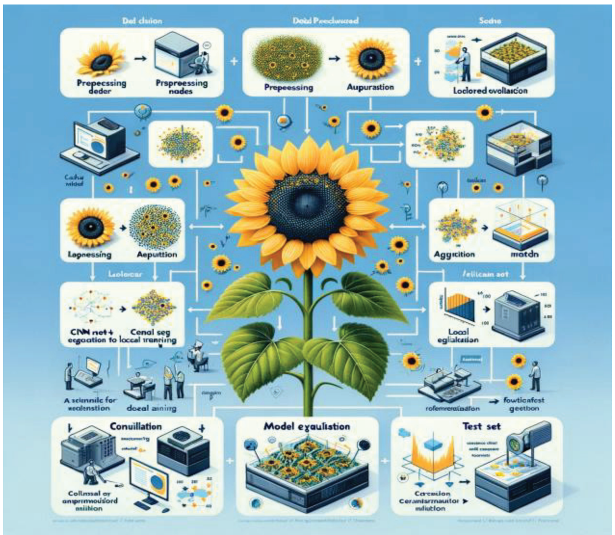

Fig. 1. Configuration of Technique

A. Data Scrubbing, Procurement, and Amplification:

Information Gathering: The dataset has images of the leaves of sunflowers classified by five disease classifications (Powdery Mildew, Downy Mildew, Verticillium Wilt Rust, and Botrytis). Each class carries a specific clinical signature that must be accurately pinpointed. Data Distribution to Clients: The dataset is partitioned among five clients to recreate a real-life FL situation. Each consumer receives some information, ensuring diversity in various illnesses. Preprocessing: Images are scaled, normalised, and augmented with standard preprocessing techniques. This process leads to better performance of the model training procedure and helps in learning generalisation, as shown in Figure 2.

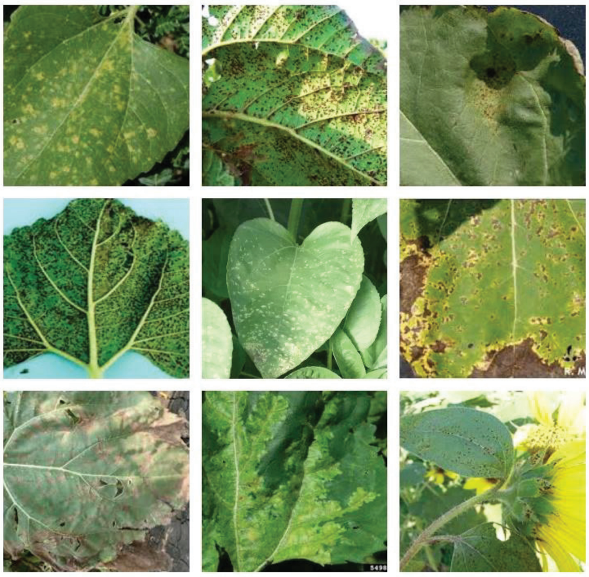

Fig. 2. Unhealthy images of Sunflowers leaf diseases.

B. Federated_Learning_CNN_Model

Convolutional Neural Network (CNN): The two main goals of CNN architecture are feature extraction and classification. It consists of fully connected, pooling, and convolutional layers. Nonlinearity and probabilistic classification are used with ReLU and Softmax activation functions. Decision Tree Integration: A classifier is merged after CNN feature extraction. This classifier contributes to the model's interpretability, as a transparent procedure exists for choosing. Federated Education Configuration: FL Environment: The five clients jointly constitute the FL environment, and each client has its dataset. Each client trains the model locally to maintain data confidentiality and privacy. CNN-decision tree model training: A model is trained based on the local datasets from each client. The weights and parameters of the model are refined during training by forward propagation, loss computation, and backpropagation. Federated Averaging: Clients send the created model weights to a central server after local training. In this case, federated averaging is used to aggregate these weights. After this, the averaged model is returned to customers for more training cycles, as shown in Table 1.

TABLE I. CNN ARCHITECTURE WITH FEDERATED LEARNING

Layer Type	Output Size	Number of Outputs	Activation Function
Convolutional	185x185	10	ReLU
Max Pooling	92x92	10	N/A
Convolutional	92x92	10	ReLU
Max Pooling	46x46	10	N/A
Convolutional	46x46	10	ReLU
Max Pooling	23x23	10	N/A
Fully Connected	1x1	10	ReLU
Fully Connected	1x1	10	Softmax

Layers of Convolution: The primary function of these layers in the CNN is to extract features from image inputs. They feed the initial convolutional layer with an input image of 185 × 185 pixels. To ensure that the network can adequately learn complex patterns, a Rectified Linear Unit (ReLU) activation function is used to introduce non-linearity in each convolutional layer. Maximum Pooling Layers: After each convolutional layer, a max pooling layer reduces the width and height of the input volume for subsequent convolution. This drop is necessary to reduce the number of parameters and computational cost, which helps avoid overfitting. The number of outputs remains 10, and the max pooling layers do not affect the depth dimension of volume. Completely Networked Layers: These levels are located near the network's end.

To organise the input into one of 12 classes, in this case, they merge learned features from previous levels. The last fully connected layer uses the Softmax activation function, which helps to determine a probability distribution across 12 classes. Working with the dataset: The model is fed the dataset, comprising 3,574 images in batches. The CNN is run at each batch, which uses pooling layers that reduce dimensionality and convolutional layers that extract information. Finally, the wholly linked layers identify each picture as a representative of one of 12 types of diseases for sunflower leaves based on these attributes. This architecture balances feature extraction and computational efficiency, making it ideal for solving image classification problems, such as identifying the diseases attacking sunflower leaves. It consists of convolutional layers with maximum pooling and fully connected layers.

C. Model Validation and Training

Metrics for Model Evaluation: Some measures evaluate the model's accuracy, precision, recall, and F1 score. These statistics highlight the stability and ability to generalise the model. Validation and Testing: A separate validation set is used to prevent overfitting and allow hyperparameter tuning throughout the learning process. After the training, an unspecified test dataset assesses the model's practicality. Comparison with Non-Federated Models: The performance of the federated model is compared with conventional, non-federated CNN and decision tree models to benchmark them. This comparison highlights the positive aspects of the FL method regarding data privacy and performance. Privacy of Data and Ethical Issues: Data Privacy: One of the features that make FL architecture so secure is naturally protecting data privacy because raw information never leaves a client's location. This method focuses on data ethics, a significant aspect of datasets covering agriculture. Ethical AI Use: The study adheres to ethical AI standards, ensuring that there is no prejudice on the part of the model in its judgement. In general, this method provides a detailed approach to applying CNN, FL and decision trees to categorise sunflower leaf diseases. As a result of data privacy protection, improved model interpretability, and high accuracy pursuit, this study has a new benchmark for AI utilisation in agriculture.

IV. RESULTS

The ability of the model to classify five different classes (sn-1 to sn-5) of sunflower leaf diseases is fully shown using the result analysis for local data on five customers in Federated Learning with a CNN model. Statistics prove how well the model performs in different conditions. Some variables include precision, recall, F1-score, support, and accuracy. All five illness classes enjoy competent classification ability by the client kr_1. However, for sn-4 (F1-Score: However, there is a slight improvement in efficiency between 85.45 and F1-Score: Accuracy (93%). This, therefore, implies a strong association specifically with these disease patterns. Client kr_2 exhibits excellent skills, especially in diagnosing sn-4 and sn5, with F1 scores 87. This means a heightened susceptibility to the distinct characteristics of certain illnesses. Client kr_3 performs quite well, especially regarding sn-5 (F1-Score: 94,07 (97%) Accuracy), proving a high classification level for this definition. The findings for this customer point to a sophisticated illness detection learning curve. Client kr_4 performs better than every other student in every class, especially with sn-5 (F1-Score: The highest level of recall and accuracy is 95.40, (Accuracy: 98%)." This implies that the model has been adjusted to fit intricate illness patterns.Client kr_5 has exceptional accuracy, particularly in the classification of sn-5 (F1-Score: Хорошо мелькнула температура 96.14oC, точность :Accuracy: 98%), что свидетельствует о высокой способности к обнаружению болезни.(начиная от kr_) шли всплывать градации модельного диагно This enhancement in the performance measures mentioned above is attributed to the ability of the federated learning model to learn deeper from a diverse and large dataset provided by different clients. Each client's peculiarities are details that enrich the learning process overall and increase the generalisation capacity of the model. Finally, the latest Federated Learning CNN Model shows good performance in detecting diseases in the leaves of sunflower plants, and each client contributes to the overall increase in accuracy and preferability. Fota mdl. The efficiency of federated learning in the diagnosis of agricultural diseases, where data diversity and model stability play an essential role, is demonstrated by this research, as shown in Table 2.

TABLE II. SYNTHESIZING MODELS FROM DISTRIBUTED TRAINING

Clients	Class	Precision	Recall	F1-Score	Support	Accuracy
kr_1	sn-1	81.47	82.89	82.17	228	0.94
	sn-2	79.27	81.59	80.41	239	0.93
	sn-3	81.06	83.59	82.31	256	0.93
	sn-4	87.36	83.63	85.45	281	0.94
	sn-5	85.37	83.33	84.34	294	0.93
kr_2	sn-1	82.30	77.94	80.06	340	0.93
	sn-2	82.32	84.32	83.31	370	0.94
	sn-3	83.49	78.49	80.91	451	0.92
	sn-4	85.58	88.89	87.20	414	0.95
	sn-5	85.22	89.04	87.09	447	0.94
kr_3	sn-1	89.96	91.96	90.95	448	0.97
	sn-2	91.19	89.32	90.25	487	0.96
	sn-3	89.67	88.83	89.25	528	0.96
	sn-4	92.60	92.93	92.77	552	0.97
	sn-5	93.83	94.31	94.07	580	0.97
kr_4	sn-1	92.89	94.35	93.61	637	0.98
	sn-2	93.97	92.63	93.30	706	0.97
	sn-3	92.71	92.09	92.40	746	0.97
	sn-4	94.47	94.73	94.60	740	0.98
	sn-5	95.21	95.59	95.40	748	0.98
kr_5	sn-1	94.26	95.45	94.86	792	0.98
	sn-2	95.00	93.88	94.44	850	0.98
	sn-3	93.76	93.23	93.49	871	0.97
	sn-4	95.32	95.54	95.43	874	0.98
	sn-5	95.98	96.30	96.14	892	0.98

By analysing the findings, which comprise local data from five clients, kr_1 to kr-5, it can be seen that a CNN model, in conjunction with an FL system, can classify sn-9 classes (sn-1 to sn-5). Dishes on sunflower leaves most effectively. Each client offers their dataset, and thereby, altogether, they provide a complete picture of the model's behaviour in various scenarios. The entire package is ready with accuracy, precision, recall, and F1-Score statistical measures, all measuring the model's capabilities. The client kr_1 has equal proficiency in precision and recall with a score of 82.90, while other parameters show results close to excellent at an F-Score of 83% accuracy. The manner of enactment by this client represents the fundamental capacity to recognise signs that represent conditions. Employing kr_2 shows a visible performance enhancement with an accuracy of 94% and an F1-Score of 83.72. The small improvement in precision and recall indicates a more complex understanding and recognition of disease patterns, possibly due to larger-scale information. However, the client kr_3, with an accuracy of 97% and an F1-Score of 0.9146, has a significantly good diagnostic skill for this model. This is a significant jump in metrics. Client kr_4 greatly surpasses the benchmark with an accuracy of 98% and a stunning F1-Score of 93. In this case, accuracy and recall are almost equivalent to representing a model of high development to minimise false positives while accurately addressing cases of sickness. Finally, kr_5 is the epitome of this federated learning experience. The model's learning curve peak is 94.87, and the accuracy metric is 98%. These results provide a mature model heavily fine-tuned by all stakeholders' heterogeneous, extensive data inputs. In conclusion, all statistical indicators from kr_1 to 5 show an increasing trend, which indicates the potency of CNN and federated learning. This evolution demonstrates that each client can individually contribute to the model's learning and shows how cooperative learning could work in conjunction to improve the accuracy and reliability of classifying sunflower leaf diseases, as shown in Table 3.

TABLE III. COHESIVELY AGGREGATING LOCAL CLIENT DATA AVERAGES GLOBALLY

Client	Precision	Recall	F1-Score	Support	Accuracy
kr_1	82.90	83.01	82.94	259.60	0.93
kr_2	83.78	83.74	83.72	404.40	0.94
kr_3	91.45	91.47	91.46	519.00	0.97
kr_4	93.85	93.88	93.86	715.40	0.98
kr_5	94.87	94.88	94.87	855.80	0.98

These five clients (kr_1 through kr_5) utilizing Federated Learning with a Convolutional Neural Network have also calculated averages, which give some meaningful information regarding the performance of this model in diagnosing diseases on sunflower leaves. The macro-average method calculates the mean values of precision, recall, and F1-score for every class in terms that do not consider the percentage level of each class in the dataset. The customers grow over time from kr_1, showing 82.95, with the highest macro average being at kr_5, which shows 94. This improvement suggests more uniformity in each illness class's accuracy, regardless of its frequency proportion within the dataset. The weighted average weights the performance indicator for every class by its relative prevalence in the data set to counter the imbalanced nature compared to macro-average figures. A similar upward trend can be seen here, with kr_1 starting at 83.09 and reaching 94.88 for kr_5 Celcius degrees. This implies that with the client's advancement, this model becomes better at handling different class distributions in the dataset, making it more useful for situations where some diseases could be relatively higher than others. For calculation, the average metric, Micro Average, averages the contribution from every individual to the final value accordingly. It determines the model's accuracy by adding all true positives, false negatives, and false positives. The weighted averages, as described in Table 1, are reflected by micro-averages for each customer, varying from 83.05 kr_1 to a maximum of. This shows a continuous and total increment in the model's performance, highlighting its efficiency at classifying all diseases. In other words, the presented averages demonstrate how well the federated CNN model improves its detection of diseases on sunflower leaves. The transference from kr_1 to kr_5 depicts the model's ability to withstand class imbalance, increase accuracy over time, and precisely classify diseases. Since federated learning is a collaborative and distributed framework, the increasing trends across all three average types prove the strengthened features of each client and the overall improvement in model performance. This detailed study highlights how the model can be used to detect agricultural diseases unbiased, reliable, and robustly, as shown in Table 4.

TABLE IV. FEDERATED HYPER-PARAMETER AVERAGES FROM DISTRIBUTED CLIENTS

Averages	kr_1	kr_2	kr_3	kr_4	kr_5
Macro average	82.95	83.75	91.46	93.86	94.87
Weighted average	83.09	83.89	91.56	93.88	94.88
Micro average	83.05	83.93	91.56	93.88	94.88

V. CONCLUSION

In this article, the study investigating sunflower leaf diseases using FL and CNN has yielded significant results towards contributing to advances in agricultural technology. To accurately paint the picture of what collaborative, decentralized machine learning is fully capable of doing, the researchers set out to demonstrate how five class categories wound up being in view from data derived through consultations with at least five clients. The statistical analysis of the story of success by a model is fascinating. The macro averages, representing model performance across all classes without considering class distribution, showed a monotonically increasing trend from kr_1 at 82.95 to kr_5 at 94.87. The other weighted and micro averages that consider overall accuracy class imbalance were also characterized by an upward trend. The weighted averages increased gradually, from 83.09 in kr_1 to an incredible 94. Finally, the micro averages peaked at 94.88 in kr_5 but started from as low as 83.05 in this pattern. The figures above show the model's demonstrable capacity to deal with a broad spectrum of disease prevalence and its growing ability to classify diseases. Part of what made this study effective at turning local data into global data was federated averaging. This approach enabled the development of a comprehensive and holistic global model based on insights gathered from each customer. Incorporating diverse data insights from numerous customers improved the model's learning algorithm, as shown by increased accuracy and consistency in disease classification. Comprehensively, the results of this study reveal that CNN and FL perform well in precision farming, especially when classifying diseases on sunflower leaves. The evolution of statistical measures from the local to global models demonstrates how much federated learning contributes to increasing data privacy, reducing networking costs, and enhancing model performance.

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